ALGEBRA 1 · UNIT 6: SYSTEMS OF EQUATIONS AND INEQUALITIES

Banquet Planner

Teacher Guide & Answer Keys

Teacher

2–3 class periods

Project Overview

Students plan a school banquet by comparing venues, DJs, and ticket prices. They solve systems by tables, graphing, substitution, and elimination to find break-even points, then recommend the best plan for different numbers of guests.

The three venues create three pairwise intersections at 40, 60, and 80 guests, which divides the number line into 'best deal' intervals. Students see why the intersection of two lines is the break-even point and how to use it to make a decision.

Standards Addressed

- A-REI.6 Solve systems of linear equations exactly and approximately
- A-REI.11 Explain why intersections are solutions
- A-CED.2 Create equations in two variables
- A-CED.3 Represent constraints and interpret solutions

Pacing (2–3 class periods)

Day	Plan
Day 1	Launch: Which venue is cheapest? Let students argue, then show it depends. Part 1 tables, equations, and graphs.
Day 2	Part 2: substitution and elimination for break-even points and purchases.
Day 3	Part 3: final plan and proposal.

Materials

- Student handout and venue cards
- Graph grid (provided)
- Colored pencils
- Calculators

How the Three Versions Differ

Level	What changes
CP2	Two venues with a table and graph, then substitution with support. Design compares two DJs.
CP1	Three venues with all break-even points, elimination with a decoration order, and a special-case (dependent) system.
Honors	Adds ticket pricing, mixed couple and single tickets, and a parameter question: what rental fee makes two venues tie at a given guest count?

Watch For (Common Misconceptions)

- Thinking the intersection is the 'best' choice rather than the point where the costs are equal.
- Setting up y = 1200x + 25 (switching rate and fixed fee).
- Stopping after finding x without finding the cost.

Extension Ideas

- Get real quotes from local venues.
- Add a capacity constraint and graph it as a vertical line.
- Add sales tax and see if the break-even point changes.

Answer Key – CP2

Part 1: Compare Two Venues

#	Task	Answer
1	Complete the table.	Grand: 1,700, 2,200, 2,700, 3,200, 3,700. Barn: 1,100, 1,800, 2,500, 3,200, 3,900.
2	At what number of guests do the two venues cost the same? How does the table show it?	80 guests: both cost $3,200 in the table.
3	Write an equation for each venue.	Grand: y = 25x + 1200. Barn: y = 35x + 400.
4	Graph both equations on the grid. Label the point where the lines cross.	Lines cross at (80, 3200).

Part 2: Solve It Algebraically

#	Task	Answer
5	Solve the system y = 25x + 1200 and y = 35x + 400 by substitution. Write the solution as an ordered pair.	25x + 1200 = 35x + 400 → 800 = 10x → x = 80; y = 25(80) + 1200 = 3200. Solution (80, 3200).
6	What does the ordered pair mean for the banquet?	With 80 guests, both venues cost $3,200.
7	If 60 guests come, which venue is cheaper and by how much?	Grand $2,700, Barn $2,500. The Barn is cheaper by $200.
8	If 100 guests come, which venue is cheaper and by how much?	Grand $3,700, Barn $3,900. The Grand Hall is cheaper by $200.

Part 3: Pick the DJ & Plan

#	Task	Answer
9	Write an equation for each DJ and solve the system to find when they cost the same.	Max: y = 2x + 300. Nova: y = 4.50x + 150. 2x + 300 = 4.50x + 150 → 150 = 2.50x → x = 60 guests ($420).
10	The council expects about 70 guests. Choose a venue and a DJ. Find the total cost.	Barn: 35(70) + 400 = $2,850. DJ Max: 2(70) + 300 = $440 (cheaper than Nova's $465). Total $3,290.
11	Write a short recommendation to the council using your break-even points.	Answers vary.

Answer Key – CP1

Part 1: Three Venues

#	Task	Answer
1	Write an equation for each venue.	Grand: y = 25x + 1200. Barn: y = 35x + 400. Sky: y = 45x.
2	Graph all three on the grid.	Three lines; Sky Lounge starts at 0 and is steepest.
3	Use substitution to find all three break-even points (one for each pair of venues).	Grand = Barn: (80, 3200). Barn = Sky: 35x + 400 = 45x → (40, 1800). Grand = Sky: 25x + 1200 = 45x → (60, 2700).
4	Use your break-even points to state which venue is cheapest for each range of guests. Remember the capacity of each venue.	Fewer than 40: Sky Lounge. 40 to 80: Riverside Barn (tie at 40 and 80). 80 to 120: Grand Hall is cheaper; above 120 only the Grand Hall fits.

Part 2: Decorations & DJs

#	Task	Answer
5	Order 1: 12 tablecloths and 30 centerpieces cost $420. Order 2: 8 tablecloths and 25 centerpieces cost $320. Use elimination to find the price of…	×2 and ×3: 24t + 60c = 840 and 24t + 75c = 960 → 15c = 120 → c = $8; 12t + 240 = 420 → t = $15
6	A committee member says: "I only remember that 12 tablecloths and 30 centerpieces cost $420, and 8 tablecloths and 20 centerpieces cost $280." Can…	No. The second equation is ⅔ of the first (multiply 8t + 20c = 280 by 1.5 to get 12t + 30c = 420). They are the same line, so infinitely many solutions; the prices can't be determined.
7	DJ Max charges $300 plus $2 per guest. DJ Nova charges $150 plus $4.50 per guest. When do they cost the same?	2x + 300 = 4.50x + 150 → x = 60 guests
8	Photo Booth A charges $250 plus $3 per guest. Photo Booth B charges $100 plus $3 per guest. Write the system. What happens when you solve it? What…	3x + 250 = 3x + 100 → 250 = 100, false. No solution: parallel lines, so Booth B is always $150 cheaper.

Part 3: Final Plan

#	Task	Answer
9	For 70 guests and for 110 guests, find the cheapest venue + DJ combination and the total cost.	70: Barn 35(70) + 400 = 2,850 + DJ Max 440 = $3,290. 110: Grand 25(110) + 1200 = 3,950 + DJ Max 2(110) + 300 = 520 → $4,470.
10	Tickets will be priced so that the banquet at least breaks even for 90 guests with your plan. What ticket price do you recommend?	90 guests: Grand 3,450 + DJ Max 480 = $3,930. 90p ≥ 3930 → p ≥ 43.67 → $44 per ticket (answers vary with plan).
11	Write a recommendation that uses at least two break-even points.	Answers vary.

Answer Key – Honors

Part 1: Three Venues

#	Task	Answer
1	Write all three cost equations and find every break-even point.	Grand $y = 25x + 1200$, Barn $y = 35x + 400$, Sky $y = 45x$. Break-even: (40, 1800) Barn-Sky; (60, 2700) Grand-Sky; (80, 3200) Grand-Barn.
2	Write the cheapest-venue rule as a piecewise description using inequalities on x (include capacity).	$0 < x < 40$: Sky. $40 \le x \le 80$: Barn (ties at endpoints). $80 < x \le 200$: Grand (the Barn holds only 120 and the Sky only 90).
3	The Barn offers to lower its rental fee to R dollars. For what value of R would the Barn and the Grand Hall tie at exactly 100 guests?	$35(100) + R = 25(100) + 1200 \rightarrow 3500 + R = 3700 \rightarrow R = \$200$
4	For what per-guest price would the Sky Lounge tie the Grand Hall at 75 guests?	$75m = 25(75) + 1200 = 3075 \rightarrow m = \$41$

Part 2: Tickets & Revenue

#	Task	Answer
5	Couple tickets cost $60 and single tickets cost $35. A total of 150 people attended and ticket sales were $4,750. How many of each ticket were sold?	$2c + s = 150$ and $60c + 35s = 4750$. $s = 150 - 2c \rightarrow 60c + 5250 - 70c = 4750 \rightarrow c = 50$, $s = 50$.
6	Show that the price per person is lower with a couple ticket. Explain how that affects revenue when more couples buy tickets.	Couple: $\$60 \div 2 = \$30$ per person < $35. Each couple ticket instead of two singles lowers revenue by $10.
7	Using the Grand Hall and DJ Max ($300 + $2 per guest), write the total cost for x guests. If every guest pays $p, write the revenue. Find the…	Cost $= 25x + 1200 + 2x + 300 = 27x + 1500$. Revenue $= 40x$. $40x = 27x + 1500 \rightarrow x = 115.4 \rightarrow 116$ guests.
8	For a single ticket price p, write the break-even attendance as a function of p. What happens as p gets close to $27?	$px = 27x + 1500 \rightarrow x = 1500 \div (p - 27)$. As p approaches $27, the denominator approaches 0, so the attendance needed grows without bound.

Part 3: Final Plan

#	Task	Answer
9	Choose the venue and DJ that minimize cost for this whole range. Justify with your break-even points.	Grand Hall + DJ Max: Grand is cheapest above 80 guests and the only venue that holds 150. DJ Max is cheaper above 60 guests.
10	Choose couple and single ticket prices so the banquet breaks even if 80 guests come as 25 couples and 30 singles. Show your system or equation.	Cost for 80 guests: 27(80) + 1500 = $3,660. Need 25C + 30S = 3660, e.g., C = $84, S = $52 (2100 + 1560 = 3660). Answers vary.
11	Write a recommendation explaining what happens to profit if 150 guests come instead.	Answers vary.

Scoring Rubric (all levels)

Criteria	4 – Exceeds	3 – Meets	2 – Approaching	1 – Beginning
Mathematical Accuracy	All calculations and solutions are correct; answers include units and are checked.	Nearly all work is correct; minor errors do not change conclusions.	Several errors; some conclusions are affected.	Many errors or missing work.
Reasoning & Justification	Clear, complete explanations connect every answer to the situation; uses precise vocabulary.	Explanations are mostly clear and connected to the situation.	Explanations are brief or partly unclear.	Little or no explanation.
Application & Design	Recommendation correctly uses the break-even points for the expected number of guests. Goes beyond the minimum with a creative, realistic choice.	Recommendation correctly uses the break-even points for the expected number of guests.	Design is incomplete or only partly uses the math from the unit.	Design is missing or unrelated to the unit math.
Communication & Presentation	Proposal shows equations, a graph with intersections labeled, the break-even analysis, and a clear recommendation. Polished, organized, and visually engaging.	Proposal shows equations, a graph with intersections labeled, the break-even analysis, and a clear recommendation. Organized and easy to follow.	Product is missing required parts or is hard to follow.	Product is mostly missing.